

Multiple Choice (10 marks)

1. Which mass is undergoing the greatest amount of acceleration?
 - (a) 1 kg subjected to a force of 1 N
 - (b) 1 kg subjected to a force of 100 N
 - (c) 100 kg subjected to a force of 1 N
 - (d) 100 kg subjected to a force of 100 N
2. Rocks sometimes show evidence of ancient glaciers. Which is the best sign that glaciers once covered an area?
 - (a) Rocks have many scratches.
 - (b) Rocks contain fossils.
 - (c) Rocks are covered by moss.
 - (d) Rocks are deep underground.
3. Planets orbit
 - (a) Earth.
 - (b) the Moon.
 - (c) the galaxy.
 - (d) the Sun.
4. Which group consists only of living things?
 - (a) rabbit, seed, bird
 - (b) seed, bird, wind
 - (c) Volcano, candle, rabbit
 - (d) wind, candle, volcano
5. An engineer is analyzing which areas in a city might become flooded if there are heavy rains. Which of the following maps is best to use for this analysis?
 - (a) a map showing the routes of city buses
 - (b) a map showing the locations of streets
 - (c) a map showing the locations of houses
 - (d) a map showing the elevations of ground surfaces

True / False (10 marks)

Answer True or False

6. True or False: A galaxy occupies the greatest amount of space compared to other astronomical objects like black holes and neutron stars.
7. True or False: Preventing sperm production in male insects can help reduce the total population of insects.
8. True or False: Pioneer plants must grow very large to successfully establish themselves on recovering beaches.
9. True or False: Reef ecosystems require adaptations to withstand extreme temperature changes similar to intertidal zones.
10. True or False: Enzymes enable reactions to occur at lower temperatures, facilitating biochemical processes in the body.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. How can Pat design an experiment to determine which plant food, "Quickgrow" or "Supergrow," promotes the best growth in a specific type of houseplant? Discuss the steps involved.
12. Discuss and analyze the properties of different types of matter, providing examples to illustrate how these properties affect their behavior and applications in everyday life.

End of Paper